BJT small-signal II

Cheat sheet · Day 12 · Wed 4 Nov · pack E7

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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SMALL-SIGNAL BJT II  (always: DC Q-point -> IE -> re = 26 mV/IE)
Reflection: R in emitter looks (b+1)R from base;  R at base looks R/b from emitter.

CE unbypassed:  Zb = b re + (b+1)RE ~ b(re+RE);  Zi = R1||R2||Zb;  Zo = RC
                Av = -RC/(re+RE) ~ -RC/RE   (no beta -> stable), 180 deg
Partial bypass: DC uses RE1+RE2;  AC uses RE1 only:  Av = -RC/(re+RE1)
Emitter foll.:  Zi = RB||b(re+RE||RL)  (RL changes Zi!)
                Zo = RE||re (ideal source);  Zo = RE||(re + (Rs||RB)/b) with Rs
                Av = RE/(RE+re) ~ 1, AvL = (RE||RL)/(RE||RL + re), in phase
Common base:    Zi = RE||re ~ re (tiny);  Zo = RC;  Av = aRC/re (in phase);  Ai = -a
Coll. feedback: Av = -(RC||RF)/re  [= -(RF/(RC+RF))RC/re];  Zo = RC||RF
                Zi = re/(1/b + RC/(RC+RF))  =  b re || RF/(1-Av)  (Miller; use AvL i
Loading:        AvL = AvNL * RL/(RL+Zo)   (two-port; exact for CE types only)
                CE: replace RC by RC||RL.   Avs = Zi/(Zi+Rs) * AvL.   AiL = -AvL*Zi/R
                |AvNL| > |AvL| > |Avs|  always
ro: ignore only if ro >= 10RC (say so). Bypassed CE: Av = -(RC||ro)/re.
Prac measurement: Zi = Rs Vi/(Vs-Vi);  Zo = (VoNL - VoL)/VoL * RL
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